Generic

cdfmvn

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CDFMVN Function

CDFMVN(*b, Sigma,* < *, mu* > **);**

The CDFMVN function enables you to compute the cumulative distribution function (CDF) of a k-dimensional multivariate normal (MVN) distribution, $k \geq 2$. Let $\mathbf{X} = (X_1, X_2, \ldots, X_k)$ be a multivariate normal random vector with mean μ and covariance Σ, which is expressed mathematically as $\mathbf{X} \sim \text{MVN}(\mu, \Sigma)$. For any k-dimensional vector, $\boldsymbol{b} = (b_1, b_2, \ldots, b_k)$, the CDF at $\boldsymbol{b}$ is the probability $P(X_1 \leq b_1, X_2 \leq b_2, \ldots, X_k \leq b_k)$. In other words, the CDF is the probability that the components of $\mathbf{X}$ are within the left-tailed region $\mathbf{X} \leq \boldsymbol{b}$. The CDF function is denoted by $\Phi_k(\boldsymbol{b}; \mu, \Sigma)$.

The CDFMVN function uses different algorithms, according to the dimension of the problem.

- When $k = 2$, the function standardizes the problem and returns a probability by calling the PROBBNRM function in Base SAS.
- When $k = 3$, the function uses the trivariate integration scheme by Genz (2004).
- When $k > 3$, the function uses a quasi-random Monte Carlo integration scheme, as described in Genz and Bretz (2009). You can use the RANDSEED subroutine to set the random number stream for the quasi-random numbers.

The arguments to the CDFMVN function are as follows:

b is an $n \times k$ matrix of upper integration limits. Each row represents an upper limit for integration for the CDF. The CDFMVN function returns an n-element column vector, where the ith element of the vector is the probability associated with the ith row of *b*.

Sigma specifies the $k \times k$ covariance matrix for the MVN distribution. The matrix must be symmetric and positive definite.

mu is an optional k-element numerical row vector that specifies the mean of the MVN distribution. If you omit this argument, the mean is assumed to be the k-dimensional zero vector.

The first example uses a specific 3×3 correlation matrix and limits from Genz and Bretz (2009, p. 4–5). Genz and Bretz state that the correct probability for this example is approximately 0.8279849.

```
/* Define limits and covariance matrix */
b = {1 4 2};
Sigma = {1.0          0.6           0.3333333333,
         0.6          1.0           0.7333333333,
         0.3333333333 0.7333333333  1.0         };
prob = cdfmvn(b, Sigma);
print prob;
```

Figure 0.1 Probability for Trivariate Normal CDF

prob
0.8279849

The second example uses a correlation matrix with a block-diagonal structure. The first variable is independent

of the other two variables. The second and third variables are correlated with each other. This implies that probability is the product of a univariate CDF in the first component and a bivariate CDF in the second and third components. The example calls the CDF and PROBBNRM functions in Base SAS to compute the univariate and bivariate probabilities, respectively.

```
/* X1 is independent of (X2, X3) */
b = {0.5 1.0 1.5};
Sigma = {1.0  0.0  0.0,
         0.0  1.0  0.5,
         0.0  0.5  1.0};
prob = cdfmvn(b, Sigma);
/* Validation: Phi(0.5) * Phi2(1.0, 1.5, 0.5) */
check = cdf("Normal", 0.5) * probbnrm(1.0, 1.5, 0.5);
print prob check;
```

Figure 0.2 Probability for Partially Independent Trivariate Normal CDF

prob	check
0.5579792	0.5579792

The third example is a five-dimensional problem with a structured covariance matrix. When the multivariate normal distribution has a zero mean, Genz and Bretz state that the correct probability for this example is approximately 0.459794. You can specify the optional *mu* argument to estimate the probability for a multivariate distribution that has a nonzero mean, as follows:

```
Sigma = {1 1 1 1 1,
         1 2 2 2 2,
         1 2 3 3 3,
         1 2 3 4 4,
         1 2 3 4 5};
b = 0:4;
prob_centered = cdfmvn(b, Sigma);

mu = {2 2 2 2 2};
prob_mu = cdfmvn(b, Sigma, mu);
print prob_centered prob_mu;
```

Figure 0.3 Probability for a Five-Dimensional Normal CDF

prob_centered	prob_mu
0.4598536	0.0198791

References

Genz, A. (2004). "Numerical Computation of Rectangular Bivariate and Trivariate Normal and *t* Probabilities." *Statistics and Computing* 14:251–260.

Genz, A., and Bretz, F. (2009). *Computation of Multivariate Normal and t Probabilities*. Dordrecht, Netherlands: Springer.